Kompetenzen

PROGRAMMING

BACKEND

FULL STACK

REACT

AI INTEGRATION

CAREER

KOMPETENZEN TECHNOLOGIES

Python Full Stack Developer with AI

125-Day Project-Driven Career Program

250 Technical Hours · 11 Phases

125 DAYS **250** TECHNICAL HOURS **02** MAIN PROJECTS

Build. Connect. Automate.

Kompetenzen Technologies

PROGRAMME OVERVIEW

Programme at a glance

A project-driven developer programme that takes you from Python fundamentals to building, securing and deploying real full-stack applications — with AI built into how you work.

125	250	2	Project-driven	02
DAYS	TECHNICAL HOURS	HOURS PER DAY	LEARNING MODEL	MAIN PROJECTS

WHAT THIS PROGRAMME SETS OUT TO DO

01 Learn to program.

02 Learn to build full-stack applications.

03 Learn to work with AI.

04 Build a professional portfolio.

05 Prepare for the developer job market.

PREREQUISITE

Basic computer skills and a genuine interest in coding. The curriculum starts from Python fundamentals.

HOW YOU LEARN

Learn, practise, build and review — every module ends in practical work.

THE IDEAL LEARNER

Who is this programme for?

This programme is suitable for students who:

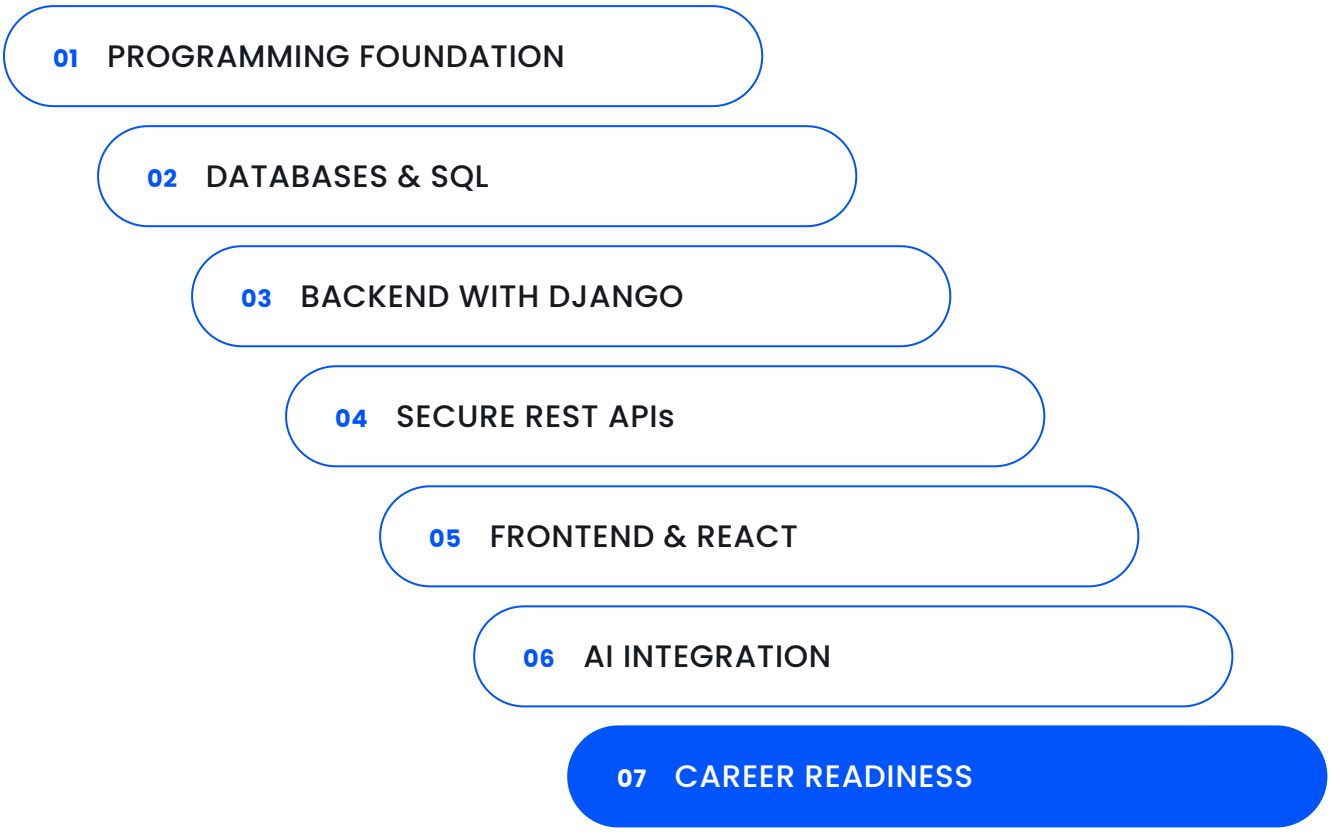
- Want to become professional software developers
- Are ready to learn programming from the fundamentals upward
- Want both backend and frontend development skills
- Want to build real applications rather than only study theory
- Want to understand how AI is changing software development
- Want a GitHub portfolio and interview preparation alongside training

PLEASE NOTE

You do not need prior professional experience. You do need commitment: 2 hours of focused training every day for 125 days, plus practice.

THE LEARNING LOGIC

Why Python Full Stack with AI?



Companies hire full stack developers who can take an idea from database to screen. This programme first teaches you how software is built end to end — and then how to use AI to build faster and add AI features to real applications.

The goal is not simply to use AI tools.
Build real software. Use AI to build it better.

WHAT YOU WILL WORK WITH

Technology stack

FRONTEND

HTML5 · CSS3 · Bootstrap · JavaScript · React

BACKEND

Python · Django · Django REST Framework · JWT · RBAC

DATABASE

MySQL · SQL

DEVELOPMENT TOOLS

Git · GitHub · VS Code · Postman

AI IN DEVELOPMENT

AI introduction · Vibe coding (AI-assisted development) · AI API integration · n8n workflow automation

GOOD TO KNOW

React is part of the core 125-day curriculum.

The frontend track covers **HTML, CSS, Bootstrap and JavaScript**, then a full **React** block (20 days).

Students integrate their React or JavaScript frontend with the Django REST backend.

11 PHASES · 125 DAYS

Your programme journey



Day 1 | Day 69 · Milestone Project 1 | Day 125 · Program end

01	ORIENTATION & INTRODUCTION	Days 1–2	4 Hrs
02	PYTHON PROGRAMMING & OOP	Days 3–32	60 Hrs
03	SQL & DATABASE ENGINEERING	Days 33–37	10 Hrs
04	DJANGO, REST API, JWT & RBAC	Days 38–63	52 Hrs
05	AI INTRODUCTION & VIBE CODING	Days 64–66	6 Hrs
06	MINI PROJECT (DJANGO)	Days 67–69	6 Hrs
07	FRONTEND FOUNDATION: HTML, CSS & JAVASCRIPT	Days 70–93	48 Hrs
08	REACT	Days 94–113	40 Hrs
09	GITHUB & PORTFOLIO	Days 114–115	4 Hrs
10	AI-INTEGRATED MAIN PROJECT	Days 116–123	16 Hrs
11	AI AUTOMATION WITH N8N	Days 124–125	4 Hrs

TOTAL **125 Days** **250 Technical Hours**

Highlighted phases end with a major project. Your batch follows either a backend-first or a frontend-first order — the content and total hours are the same.

PHASE 01

01 Orientation & Introduction

Days 1–2 | **4** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11

Day 1 · Day 69 · Milestone Project 1 · Day 125 · Program end

WHAT YOU WILL LEARN

You will understand how the programme works, what you will build and how the learning route is organised.

KEY TOPICS

- Programme orientation
- Learning route
- Technical orientation
- Course introduction

PRACTICAL OUTCOMES

01 Clear programme roadmap

02 Understanding of the learning route

PHASE 02

02 Python Programming & OOP

Days 3–32 | **60** HOURS

01 | **02** | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11

Day 1 — Day 69 · Milestone Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will learn Python from first principles to object-oriented programming, file handling and connecting Python to MySQL.

KEY TOPICS

Introduction & applications · Features · Installation · print() & input()
Data types & methods · Operators · Conditions · Looping · Functions · Scope & LEGB
Lambda · Recursion · Strings · Lists · Tuples · Sets · Dictionaries · Assessment
GitHub basics · Repositories · import & dunder · Built-in modules · Read & write
with statement · seek & tell · Binary files · Serialization · Logical programs
Pattern programs · Class & object · self · Access modifiers · Inheritance & MRO
Polymorphism · Overloading & overriding · Getters & setters · Abstraction · Encapsulation
Magic methods · Composition · Exception handling · Functional programming
MySQL connection · CRUD · SQL injection · Sample resume sharing

PRACTICAL OUTCOMES

01 Python programming exercises

02 Data-processing programs

03 OOP applications

04 Database-backed Python application

PHASE 03

03 SQL & Database Engineering

Days 33–37 | **10** HOURS

01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11

Day 1 · Day 69 · Milestone Project 1 · Day 125 · Program end

WHAT YOU WILL LEARN

You will design databases, apply constraints and write queries — from basic commands to joins and normalisation.

KEY TOPICS

- Data integrity
- SQL operators
- Database creation
- SQL constraints
- SQL commands
- Data manipulation
- Aggregate functions
- ORDER BY
- GROUP BY
- ALTER TABLE
- Wildcard characters
- JOINS
- Normalisation
- SQL practice

PRACTICAL OUTCOMES

01 Database schema

02 SQL exercise set

03 Multi-table queries

04 Normalised design

PHASE 04

04 Django, REST API, JWT & RBAC

Days 38–63 | **52** HOURS



WHAT YOU WILL LEARN

You will build Django web applications, then REST APIs with authentication and role-based access, and connect them to a frontend.

KEY TOPICS

Web fundamentals | MVT architecture | URLs | Views | HttpResponse | render() | Templates | Template inheritance | Models | ORM with MySQL | ORM relationships | Forms (GET & POST) | CSRF token | ModelForm | Custom validation | Todo form | CBV | REST concepts | REST CRUD | Default routing | Pagination | Searching | Filtering | JWT token generation | Role-based access control | Permissions | Backend CRUD design | Frontend CRUD design | API integration | External Saturday review

PRACTICAL OUTCOMES

- **01** First Django application
- **02** CRUD web application
- **03** JWT-secured REST API
- **04** API-integrated CRUD application

PHASE 05

05 AI Introduction & Vibe Coding

Days 64–66 | **6** HOURS

01 | 02 | 03 | 04 | **05** | 06 | 07 | 08 | 09 | 10 | 11

Day 1 — Day 69 · Milestone Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will learn the basics of AI and how to use AI-assisted coding tools productively — without losing understanding of your own code.

KEY TOPICS

- AI fundamentals
- AI in software development
- AI-assisted coding
- Prompting for code

PRACTICAL OUTCOMES

01 AI concepts overview

02 AI-assisted build exercise

PHASE 06

06 Mini Project (Django)

Days 67–69 | **6** HOURS

01 | 02 | 03 | 04 | 05 | **06** | 07 | 08 | 09 | 10 | 11

Day 1 — Day 69 · Milestone Project 1 — Day 125 · Program end

WHAT YOU WILL LEARN

You will build a Django mini project that brings together everything from the backend phases.

KEY TOPICS

Python | MySQL | Django | ORM | Forms | REST | JWT | RBAC | Mock interview (Day 69)

PRACTICAL OUTCOMES

01 Working Django mini project

02 Mock interview experience

PHASE 07

07 Frontend Foundation: HTML, CSS & JavaScript

Days 70–93 | **48** HOURS



WHAT YOU WILL LEARN

You will learn how web pages are structured, styled and made interactive with JavaScript.

KEY TOPICS

HTML structure | Text formatting | CSS types & properties | Tables & lists | HTML forms | Position properties | Flexbox | Grid | Semantic elements | Bootstrap | Webpage build | Variables | Data types & operators | Type coercion | Conditionals & switch | Loops | Functions | Strings & arrays | Objects | DOM | Events | JS OOP | Asynchronous functions | Error handling | JSON data handling

PRACTICAL OUTCOMES

01 Static website

02 Responsive Bootstrap site

03 Interactive JavaScript pages

04 Async JSON application

PHASE 08

08 React

Days 94–113 | **40** HOURS



Day 1 | Day 69 · Milestone Project 1 | Day 125 · Program end

WHAT YOU WILL LEARN

You will build modern React applications — components, hooks, routing, data fetching and CRUD.

KEY TOPICS

React intro · Components & JSX · Props & state · Event handling · Styling in React · React in depth · Rendering & lifecycle · useState · useEffect · useRef · Custom hooks · Conditional rendering · Lists · React forms · Prop drilling & useContext · React Router · useNavigate · useLocation · Lazy loading · Error boundaries · Data fetching · Local storage · CRUD · CRUD with local storage

PRACTICAL OUTCOMES

01 React component app

02 Routed single-page app

03 React CRUD application

PHASE 09

09 GitHub & Portfolio

Days 114–115 | **4** HOURS



WHAT YOU WILL LEARN

You will put your projects on GitHub and organise them as a professional portfolio.

KEY TOPICS

- Version control
- Repositories
- Portfolio organisation

PRACTICAL OUTCOME

01 GitHub portfolio

PHASE 10

10 AI-Integrated Main Project

Days 116–123 | **16** HOURS



WHAT YOU WILL LEARN

You will build your main project: a full-stack application with AI features integrated through APIs.

KEY TOPICS

Django | Django REST Framework | MySQL | JWT | React / JavaScript | AI API integration | GitHub

PRACTICAL OUTCOMES

01 Full-stack AI-integrated application

02 Project on GitHub

PHASE 11

11 AI Automation with n8n

Days 124–125 | **4** HOURS



Day 1 | Day 69 · Milestone Project 1 | Day 125 · Program end

WHAT YOU WILL LEARN

You will build automated workflows that connect apps and AI using n8n.

KEY TOPICS

n8n | Workflow automation | AI in workflows

FINAL OUTCOMES

01 Automated AI workflow

DEMONSTRABLE PROJECT WORK

Two major projects

Days 67–69

MILESTONE PROJECT 1

Django Mini Project

YOU DEMONSTRATE

Database design · CRUD · Templates · REST APIs · Authentication · Role-based access · Python · MySQL · Django · Django ORM

Days 116–123

MAIN PROJECT

API-Integrated Full-Stack Application with AI

YOU DEMONSTRATE

Backend API · Database · Authentication · React / JS frontend · API integration · AI features · Code on GitHub · Django · Django REST Framework · MySQL · JWT

Students do not finish with only certificates.
They finish with demonstrable project work.

CHECKING YOUR PROGRESS

Reviews & interview readiness

Your progress is reviewed at the end of every major module, and you will face a mock interview before placement.

CHECKPOINT	WHEN	COVERS
Internal review	**Day 13**	Python fundamentals
External review	**Day 14**	Python fundamentals
Internal review	**Day 31**	Python complete
External review	**Day 32**	Python complete
Internal review	**Day 63**	Django & REST (plus external Saturday review)
Mock interview	**Day 69**	Right after the mini project
Review	**Day 93**	HTML, CSS & JavaScript
Review	**Day 113**	React
Final review	**Day 125**	Whole programme & portfolio

REVIEWS

Trainers check your understanding, your code and your project progress — and tell you exactly what to improve.

MOCK INTERVIEW

Practise a real technical and HR-style interview before you apply for jobs.

PLACEMENT SUPPORT AND CAREER ASSISTANCE

Kompetenzen digital career & placement ecosystem

As a student, you can access the Kompetenzen digital placement ecosystem at www.kompetenzen.in for:

01
Job discovery
Students can explore job opportunities listed on the Kompetenzen platform.

02
Direct application
Students can apply directly to suitable job opportunities.

03
ATS resume
Students can create and maintain an ATS-friendly resume.

04
Career profile
Students can maintain their professional profile for placement opportunities.

05
Consultant support
Students can contact Kompetenzen career/placement consultants for guidance and assistance.

06
Interview readiness
Reviews and the mock interview prepare students for applications and interviews.

These are placement support and career assistance services. They do not constitute a guarantee of employment.

YOUR PORTFOLIO PROGRESSION

What you will be able to build

- 01 PYTHON PROGRAMS & OOP APPLICATION
- 02 DATABASE-BACKED PYTHON APP
- 03 DJANGO CRUD APPLICATION
- 04 JWT-SECURED REST API
- 05 DJANGO MINI PROJECT
- 06 RESPONSIVE BOOTSTRAP WEBSITE
- 07 INTERACTIVE JAVASCRIPT APP
- 08 REACT CRUD APPLICATION
- 09 AI-INTEGRATED MAIN PROJECT
- 10 N8N AI WORKFLOW

FINAL STUDENT OUTCOME

> By the end of the programme, students will have developed the skills to build full-stack applications using Python, Django and React, work with databases and REST APIs, implement JWT authentication and role-based access, use AI-assisted development, integrate AI into applications, automate workflows with n8n and present their work professionally.

Build. **Connect.** **Automate.** **Present.**